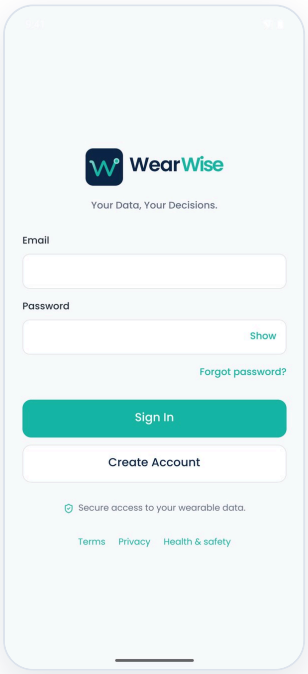
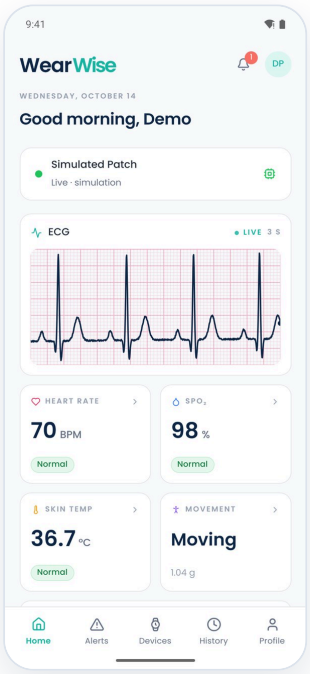
WearWise

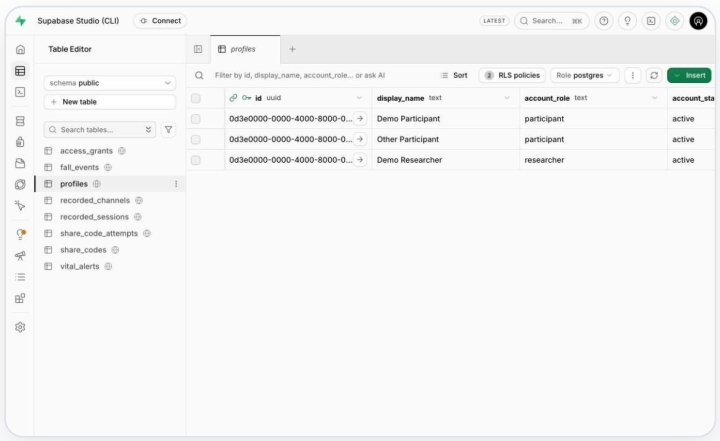
SETUP GUIDE

Run WearWise on your own computer

Install the tools, then start the backend, the web dashboard and the phone app, and set up a patch. Written for people with little or no development experience.







Handover release v1.0.0 · app 0.1.0 · firmware 0.9.6

October 2026 · WearWise Project Team

Zakiul Atheef · Hari Bethelli · Siri Bethelli · Thevindu Kodagoda Munasinghage

Research prototype.
Not a medical device.

Contents

PART 1 · BEFORE YOU START

1 What you'll end up with 3
2 How to use this guide (the terminal) 4
3 What you need 4

PART 2 · INSTALL THE TOOLS

4 Git (downloads the code) 5
5 Node.js (runs the dashboard and builds the app) 5
6 Docker Desktop (runs the backend) 5
7 Supabase CLI (controls the backend) 6
8 Android Studio and Java (builds the phone app) 6
9 Check everything 7

PART 3 · GET WEARWISE RUNNING

10 Download the code 8
11 Start the local backend 8
12 Look inside the database (Studio) 10
13 Connect the apps to your backend 11
14 Start the web dashboard 12
15 Emails on your computer (Mailpit) 13

PART 4 · RUN THE PHONE APP

16 On the Android emulator 14
17 On a real Android phone 15
18 On an iPhone (Mac only) 16
19 Try the whole system 16

PART 5 · THE WEARWISE PATCH

20 Parts and wiring 18
21 Install the firmware (Patch setup) 19
22 Pair the patch with a phone 21
23 Optional: build the firmware from source 21

PART 6 · EVERYDAY USE AND NEXT STEPS

24 Daily start and stop 22
25 Moving to a hosted backend 22
26 Running the tests 23
27 Troubleshooting 23
28 Glossary 24

PART 1

Before you start

What you'll build, what you need, and how to type commands.

1 What you'll end up with

By the end of this guide, the whole WearWise system will be running on your own computer, with no accounts or services from the original developers:

1

A local backend

The database, logins and file storage, running inside Docker on your computer, filled with safe **demo data**.

2

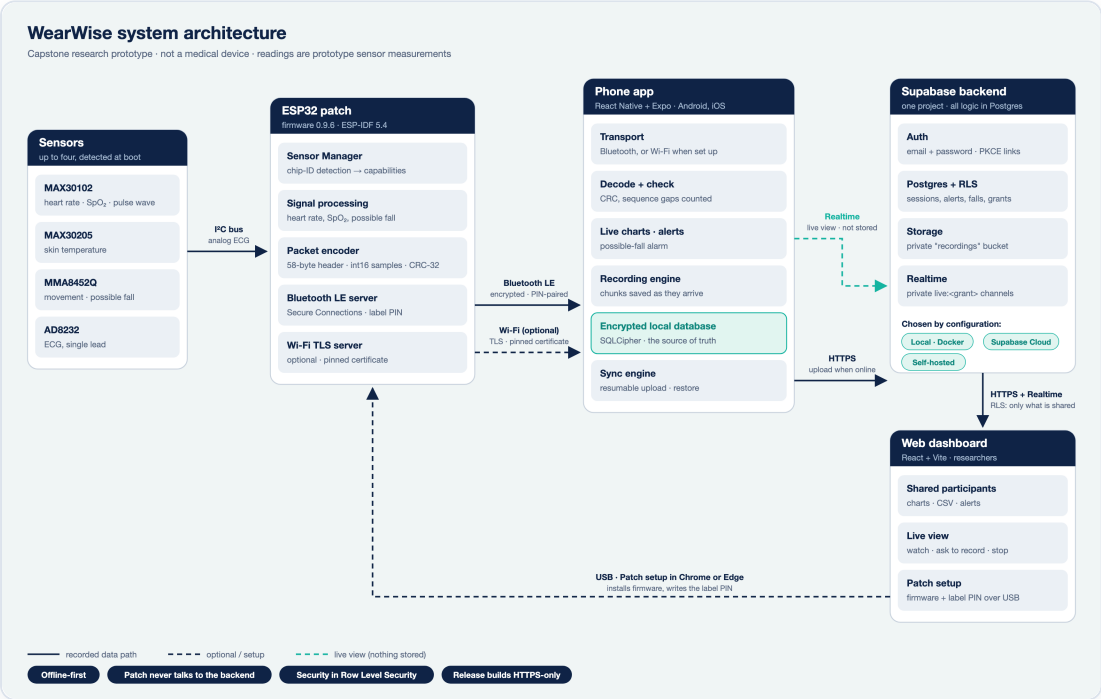
The web dashboard

Running in your browser at `localhost:5192`. You'll sign in as a demo researcher.

3

The phone app

On an Android emulator (a virtual phone on your computer) or a real phone, talking to your local backend.



The WearWise system. In this guide the "Supabase backend" box runs on your computer. A real patch is optional: the app has a built-in simulated patch.

Time needed	About **2–3 hours** the first time, mostly waiting for downloads. Later, starting everything takes about 2 minutes.
Disk space	About **25 GB free**: Docker images (several GB), Android Studio and an emulator (about 15 GB), and the project (about 2 GB).

Computer	macOS, Windows 10/11 or Linux, with administrator rights, 16 GB of memory recommended, and an internet connection.

2 How to use this guide (the terminal)

Most steps are commands you type into a **terminal**: a window where you type instructions instead of clicking. That's all a terminal is.

Your computer	How to open a terminal
macOS	Press `⌘ Space`, type **Terminal** and press Enter.
Windows	Click Start, type **Terminal** (or **PowerShell**) and press Enter.
Linux	Press `Ctrl Alt T`.

In this guide, commands look like this. Type the text after the **$** (not the $ itself) and press **Enter**. Copying and pasting is fine.

Terminal: what you type, and what you should see

```
$ node --version
v24.19.0
```

- Wait until the command finishes and the cursor comes back before typing the next one.
- Some commands keep running (the dashboard and the app's bundler). Leave that window open, and open a **new terminal window** for the next command.
- To stop a running command, click its window and press `Ctrl C`.
- Text in angle brackets is something you replace, for example `<repository-url>`.
- `cd` means "change directory" (go into a folder). Most commands must be run from the **wearwise** folder that you create in section 10.

> **Screenshots and output**
> The terminal output in this guide was captured on a real setup. Keys and private network addresses are hidden or replaced with examples, so your numbers will differ.

3 What you need

- ✓ A computer as described in section 1.
- ✓ **Access to the WearWise code repository**, for example an invitation to the project's GitHub repository.
- ✓ **Google Chrome or Microsoft Edge**. Patch setup needs one of them.
- ✓ *Optional:* an **Android phone** with a USB cable, to use a real patch over Bluetooth.
- ✓ *Optional:* the **patch hardware**: an ESP32 board and sensors (Part 5). Without a phone or patch, the emulator and the simulated patch drive the whole system.

PART 2

Install the tools

Six free programs, one at a time. After each one, a quick check proves it worked.

4 Git (downloads the code)

- **macOS:** in a terminal, run `xcode-select --install` and click **Install**, or download it from **git-scm.com**.
- **Windows:** download the installer from **git-scm.com** and accept the default options.
- **Linux (Ubuntu/Debian):** `sudo apt install git`.

Check

```
$ git --version
git version 2.54.0
```

5 Node.js (runs the dashboard and builds the app)

Go to **nodejs.org** and download the **LTS** version (22 or newer). Run the installer with the default options. npm, Node's package installer, comes with it.

Check (open a new terminal first)

```
$ node --version
v24.19.0
$ npm --version
11.17.0
```

6 Docker Desktop (runs the backend)

1. Download **Docker Desktop** from **docker.com** and install it. On Windows, accept the offer to install or enable **WSL 2**, and restart if asked.
2. Open Docker Desktop and accept its terms. Wait until it says **Engine running** (bottom left).
3. Docker must be running whenever you use the local backend. You can set it to start when you sign in (Settings → General).

Check (open a new terminal first)

```
$ docker --version
Docker version 29.8.2, build 7fc2dff
```

> **"docker: command not found" on a Mac?**
> Close the terminal and open a new one. If it still happens, open Docker Desktop → Settings → Advanced and choose to install the command-line tools system-wide.

7 Supabase CLI (controls the backend)

The Supabase CLI starts, stops and resets the local backend. Install it **as a separate program**:

System	How to install
macOS	First install **Homebrew** (follow the one-line command on **brew.sh**), then run `brew install supabase/tap/supabase`
Windows	First install **Scoop** (follow the commands on **scoop.sh** in PowerShell), then run `scoop bucket add supabase https://github.com/supabase/scoop-bucket.git` `scoop install supabase`
Linux	Download the package for your system from **github.com/supabase/cli/releases** and install it (for example `sudo dpkg -i supabase_*.deb`).

Check: 2.120 or newer

```
$ supabase --version
2.120.0
```

> **Don't use "npx supabase"**
> The WearWise project blocks npm packages from running install scripts, for security, so the npm version of the Supabase CLI can't download its program. Always use the separately installed `supabase` command.

8 Android Studio and Java (builds the phone app)

You need these to build and run the phone app, on an emulator or a real Android phone.

1. Download **Android Studio** from **developer.android.com/studio** and install it. Open it and choose the **Standard** setup. It downloads the Android SDK and an emulator, which takes a while.
2. In Android Studio: **More Actions → SDK Manager**. Under *SDK Platforms*, tick **Android 16 (API 36)**. Under *SDK Tools*, make sure **Android SDK Build-Tools**, **Platform-Tools** and **Emulator** are ticked. Click **Apply**.
3. **More Actions → Virtual Device Manager → +** (Create device). Pick a phone such as **Pixel 8**, then a system image with **API 36** (click the download icon next to it), then **Finish**.
4. Install **Java JDK 17**, the version WearWise was tested with. For example, use **Eclipse Temurin 17** from **adoptium.net** (choose version 17 and the installer for your system).
5. Tell your terminal where Android and Java are (below), then open a new terminal.

macOS: add these lines to the file ~/.zshrc

Run `open -e ~/.zshrc` (create the file with `touch ~/.zshrc` if it doesn't exist), paste the lines at the end, save, and open a new terminal.

~/.zshrc

```
export ANDROID_HOME="$HOME/Library/Android/sdk"
export PATH="$PATH:$ANDROID_HOME/platform-tools:$ANDROID_HOME/emulator"
export JAVA_HOME="$(/usr/libexec/java_home -v 17)"
```

Windows: set environment variables

Start → type **environment variables** → *Edit the system environment variables* → **Environment Variables…**. Under *User variables*:

- **New…** `ANDROID_HOME` = `C:\Users\<you>\AppData\Local\Android\Sdk`
- **New…** `JAVA_HOME` = the folder the JDK 17 installer created, for example `C:\Program Files\Eclipse Adoptium\jdk-17…`
- Select **Path → Edit → New** and add `%ANDROID_HOME%\platform-tools`.

Linux: add `export ANDROID_HOME=$HOME/Android/Sdk`, the same PATH line as macOS, and `JAVA_HOME` pointing at your JDK 17 to `~/.bashrc`.

Check (new terminal)

```
$ adb version
Android Debug Bridge version 1.0.41
Version 37.0.1-15733141
$ java -version
openjdk version "17.0.20.1" 2026-08-18 LTS
```

Building for iPhone needs a Mac with **Xcode** (from the App Store) and **CocoaPods**. See section 18.

9 Check everything

Open a **new** terminal and run these five commands. Each should print a version number. If one says "command not found", go back to its section.

Terminal

```
$ git --version
git version 2.54.0 (Apple Git-157)
$ node --version
v24.19.0
$ npm --version
11.17.0
$ docker --version
Docker version 29.8.2, build 7fc2dff
$ supabase --version
2.120.0
```

These are the versions WearWise was tested with. Newer versions are normally fine. Node must be 22 or newer, and the Supabase CLI 2.120 or newer.

PART 3

Get WearWise running

Download the code, start the backend, connect the apps, and open the dashboard.

10 Download the code

Pick a place for the project, such as your Documents folder, then download ("clone") it and install its building blocks:

Terminal

```
$ cd ~/Documents
$ git clone <repository-url> wearwise
$ cd wearwise
$ npm install
(the first time it says "added … packages" and takes a few minutes)
up to date, audited 1176 packages in 7s

199 packages are looking for funding
  run `npm fund` for details

69 vulnerabilities (16 moderate, 52 high, 1 critical)
…
```

- `<repository-url>` is the address of the WearWise repository, for example `https://github.com/<organisation>/wearwise.git`. For a private repository, Git asks you to sign in to GitHub the first time. **GitHub Desktop** (*File → Clone repository*) also works if you prefer clicking.
- `npm install` takes a few minutes the first time. On later runs it says *up to date*.

About the "vulnerabilities" message

npm reports known advisories in the project's development tools (Expo CLI, Metro, Jest). This doesn't stop anything working, and the handover documents record it (`docs/knowledge-transfer.md`). **Don't run** `npm audit fix --force`: it downgrades Expo and breaks the app. Fixing them properly needs an Expo upgrade.

11 Start the local backend

Make sure Docker Desktop is running, then, in the **wearwise** folder:

Terminal

```
$ supabase start
```

The first time, this downloads the Supabase images (several GB), which can take 5–20 minutes. Then it creates a fresh database, applies the 11 WearWise **migrations** (the steps that build the tables and security rules), loads the **demo data** and uploads the demo recordings. It finishes by printing a summary like this, which you can show again at any time with `supabase status`:

supabase status (keys hidden in this guide)

```
$ supabase status
supabase local development setup is running.

┌──────────────────────────────────────────┐
│ 🔧 Development Tools                     │
├────────────┬─────────────────────────────┤
│ Studio     │ http://127.0.0.1:54323      │
│ Mailpit    │ http://127.0.0.1:54324      │
│ MCP        │ http://127.0.0.1:54321/mcp  │
└────────────┴─────────────────────────────┘

┌──────────────────────────────────────────────────────┐
│ 🌐 APIs                                              │
├──────────────────┬───────────────────────────────────┤
│ Project URL      │ http://127.0.0.1:54321            │
│ REST             │ http://127.0.0.1:54321/rest/v1    │
│ GraphQL          │ http://127.0.0.1:54321/graphql/v1 │
│ Edge Functions   │ http://127.0.0.1:54321/functions/v1 │
└──────────────────┴───────────────────────────────────┘

┌──────────────────────────────────────────────────────┐
│ 🔑 Authentication Keys                               │
├──────────────┬───────────────────────────────────────┤
│ Publishable  │ ·········· (hidden)                   │
│ Secret       │ ·········· (hidden)                   │
└──────────────┴───────────────────────────────────────┘
… (Database and Storage sections)
```

Project URL `:54321`	The address the apps talk to.
Studio `:54323`	A website for looking inside the database (section 12).
Mailpit `:54324`	Catches every email the local backend "sends" (section 15).
Keys	You never need to copy them by hand: section 13 does it for you. The **Secret** key must never go into an app or a document.

Start again from clean demo data

`supabase db reset` wipes the local database and rebuilds it from the migrations and demo data. It's handy if you've made a mess while testing. It never touches anything outside your computer.

Terminal

```
$ supabase db reset
Resetting local database...
Recreating database...
Initialising schema...
Seeding globals from roles.sql...
Applying migration 20260826000001_profiles.sql...
Applying migration 20260826000002_recordings.sql...
… (9 more migrations)
Applying migration 20261017000011_terms_2026_10_17.sql...
Seeding data from supabase/seed.sql...
Seeding data from supabase/seed/demo-recordings.sql...
Restarting containers...
Updating Storage bucket: recordings
Uploading: supabase/seed/recordings/… (9 demo recording files)
Finished supabase db reset on branch main.
```

12 Look inside the database (Studio)

Open **http://127.0.0.1:54323** in your browser. This is **Supabase Studio**, a control panel for your local backend.

Authentication → Users (the padlock icon on the left) lists the three demo accounts.

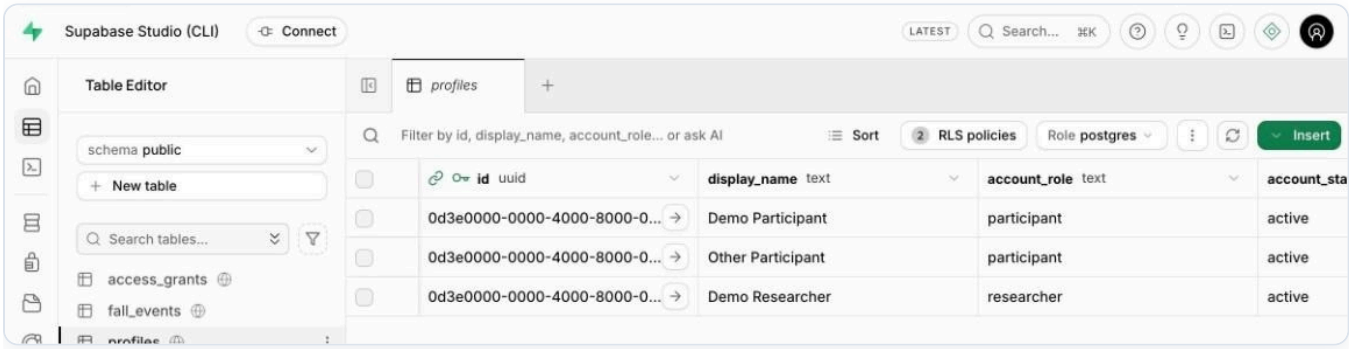


Table Editor → profiles (the table icon on the left). `recorded_sessions` holds 3 simulated demo sessions.

The demo accounts (local only; every health value is simulated)

Email	Password	What it is
`demo.participant@example.invalid`	`wearwise-demo-only`	A participant with simulated recordings, shared with the researcher. Use in the app.
`demo.other@example.invalid`	`wearwise-demo-only`	A participant who shares nothing.
`demo.researcher@example.invalid`	`wearwise-demo-only`	A researcher. Use on the dashboard.

Creating a researcher account

Researcher and clinician accounts aren't created in the app. In Studio, open the **SQL Editor**, paste the file `supabase/admin/create-viewer.sql`, change the three values marked `<-- CHANGE` (email, role, name), and click **Run**. The result shows a one-time password: give it to the person privately. To turn an existing account into a researcher, use `supabase/admin/set-role.sql` the same way.

13 Connect the apps to your backend

The dashboard and the app need to know the backend's address and its public key. One command finds both and writes them into each app's settings file:

Terminal (in the wearwise folder)

```
$ npm run backend:local

> backend:local
> node scripts/use-backend.mjs local

Mobile app (development builds) → http://192.168.1.20:54321
Dashboard (vite dev)            → http://127.0.0.1:54321

Restart Metro with a clean cache so the app picks it up:  cd apps/mobile && npx expo start -c
A phone must be on the same network as this computer, and the computer's firewall must allow port
54321.
```

Why two addresses? On a phone, "127.0.0.1" means the phone itself. So the app gets your computer's address on your Wi-Fi network (here `192.168.1.20`; yours will differ), and the dashboard, which runs on the computer, uses `127.0.0.1`.

Check what each app will use at any time:

Terminal

```
$ npm run backend:status

Mobile app, development builds: http://192.168.1.20:54321  (from apps/mobile/.env.development.local)
Mobile app, release builds:     https://<your-project>.supabase.co  (from
apps/mobile/.env.production)
Dashboard, vite dev:            http://127.0.0.1:54321  (from apps/dashboard/.env.development.local)
Dashboard, production build:    https://<your-project>.supabase.co  (from apps/dashboard/.env)
```

If you use…	Then
The Android emulator	The default usually works. If the app can't reach the backend, run `npm run backend:local -- --host 10.0.2.2` (the emulator's name for your computer).
A real phone	Phone and computer on the **same Wi-Fi** (not a guest network). When your computer asks whether to allow incoming connections (macOS) or shows a Windows Defender Firewall prompt, click **Allow** (on Windows, for *Private networks*).

If you use…	Then
A different Wi-Fi network later	Your computer's address changes. Run `npm run backend:local` again and restart the app's bundler with `-c` (section 16).

These settings files (`.env.development.local`) are only used by development builds, so they can never leak into a release build. They're also excluded from Git. `npm run backend:cloud` switches back to the hosted backend (section 25).

14 Start the web dashboard

Open a **new terminal window** (the dashboard keeps running in it), go to the wearwise folder and start it:

Terminal 2: leave it running

```
$ cd ~/Documents/wearwise
$ npm run dev -w dashboard

> dashboard@0.1.0 dev
> vite

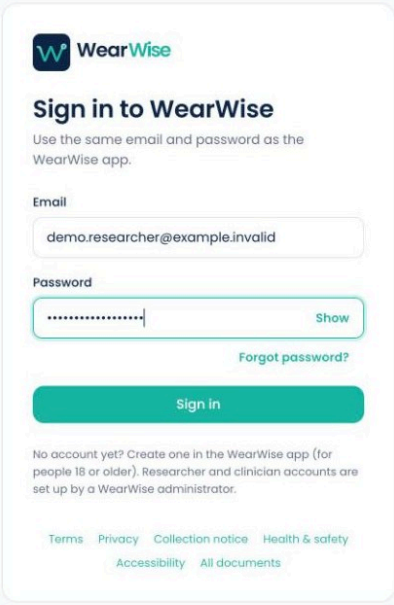
  VITE v8.3.4  ready in 107 ms

  ➜  Local:   http://localhost:5192/
  ➜  Network: use --host to expose
```

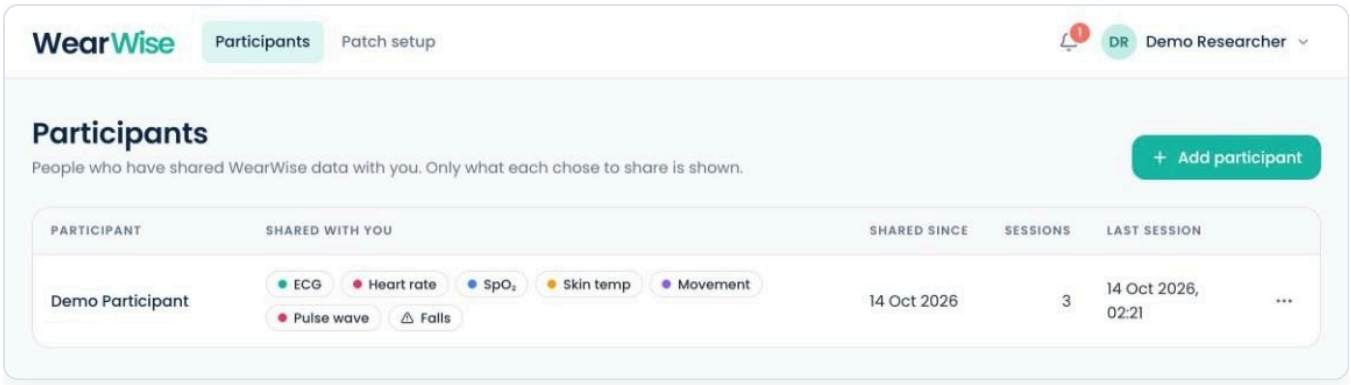
1. Open **http://localhost:5192** in Chrome or Edge.
2. Sign in as `demo.researcher@example.invalid` with the password `wearwise-demo-only`.
3. **Participants** lists *Demo Participant*. Click it to see the simulated sessions, charts, a high-heart-rate alert and a possible fall, and to download CSV files.

The **WearWise User Guide** explains every dashboard screen.

To stop the dashboard, press `Ctrl C` in its window.



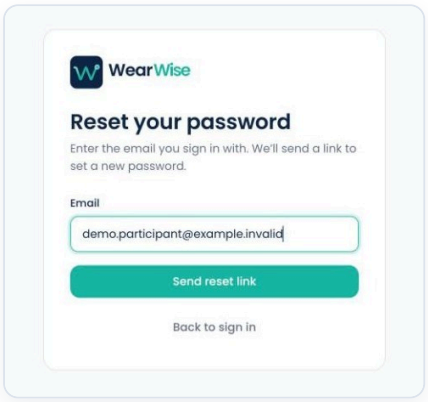
The dashboard sign-in page, served from your computer.



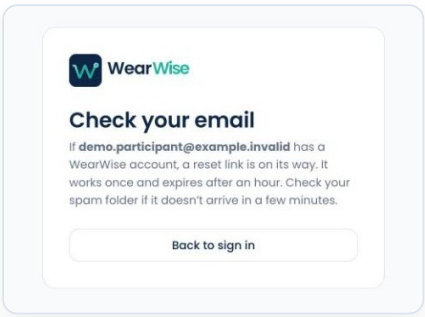
Signed in as the demo researcher. If you see this, the backend and the dashboard both work.

15 Emails on your computer (Mailpit)

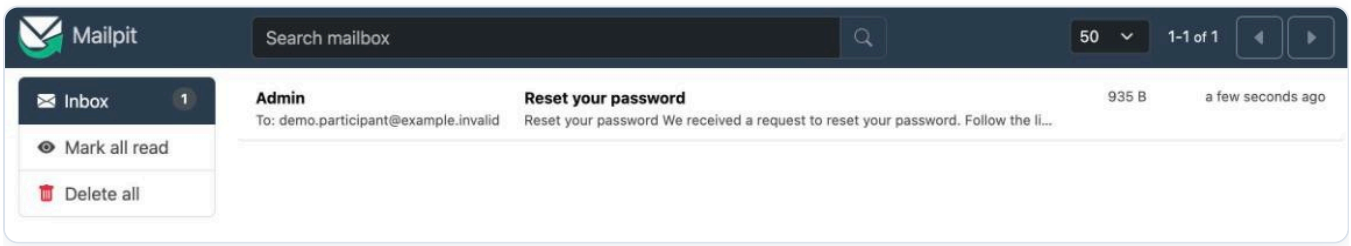
The local backend doesn't send real emails. Every email, such as password resets and sign-up confirmations, lands in **Mailpit** at **http://127.0.0.1:54324**. Try it:



1. Dashboard → *Forgot password?* → enter a demo email → *Send reset link.*



2. The dashboard confirms. Nothing has left your computer.



3. Mailpit shows the email. Click it to read it and use its link.

Locally, sign-up email confirmation is switched off, so new app accounts are signed straight in.

PART 4

Run the phone app

Build the app onto an Android emulator or a real phone, then try the whole system.

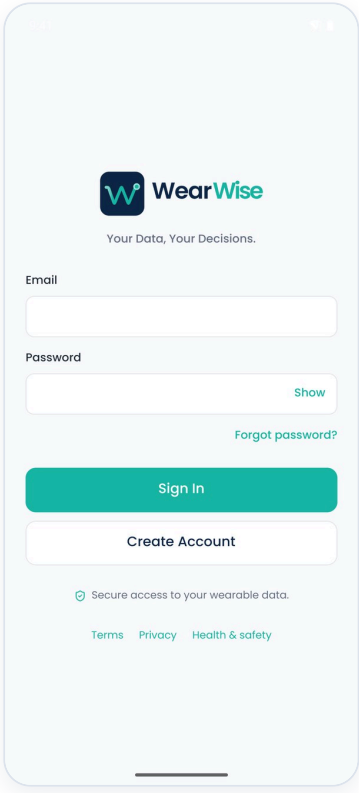
16 On the Android emulator

1. In Android Studio, open **Virtual Device Manager** and click ▶ next to your virtual phone. Wait until its home screen appears.
2. Open a **new terminal window** and build the app:

Terminal 3: leave it running

```
$ cd ~/Documents/wearwise/apps/mobile
$ npx expo run:android
… builds the app (5–15 minutes the first time), installs it on the emulator and opens it …
```

3. WearWise opens on the **sign-in screen**, with no "backend not configured" message. Sign in as `demo.participant@example.invalid` / `wearwise-demo-only`. The demo recordings are restored into **History**.
4. The terminal keeps running **Metro**, the "bundler" that sends the app its code. Leave it open while you use the app.



The app's first screen.

Next time

You only rebuild after changing native settings or updating packages. Normally, start the emulator, open the WearWise app on it, and start just Metro:

Terminal 3

```
$ npx expo start -c
Starting project at …/wearwise/apps/mobile
Starting Metro Bundler
warning: Bundler cache is empty, rebuilding (this may take a minute)
Waiting on http://localhost:8081
Logs for your project will appear below.
Android Bundled 3709ms node_modules/expo-router/entry.js (2346 modules)
```

`-c` clears Metro's cache. Always use it after `backend:local` or `backend:cloud`, because the backend address is built into the bundle. In the Metro window, press **r** to reload the app.

Expo Go won't work

WearWise uses its own native code (Bluetooth, encrypted storage), so it always runs as its own "development build" made by `npx expo run:android`. The Expo Go app from the Play Store can't open it.

17 On a real Android phone

A real phone is needed to use a real patch, because emulators have no Bluetooth.

1. **Turn on Developer options:** on the phone, open Settings → **About phone** (on Samsung: About phone → **Software information**) and tap **Build number** 7 times.
2. **Turn on USB debugging:** Settings → **Developer options** → **USB debugging** on.
3. Connect the phone with a USB cable. On the phone, tap **Allow** when it asks to allow USB debugging from this computer.
4. Check the computer can see it:

Terminal

```
$ adb devices
List of devices attached
R3C…(your phone's serial)       device
emulator-5554   device
```

5. Put the phone on the **same Wi-Fi** as the computer, then build and install:

Terminal (apps/mobile)

```
$ npx expo run:android --device
… choose your phone from the list …
```

6. If the list says *unauthorized*, unlock the phone and accept the USB debugging prompt.

Phone not on the same Wi-Fi?

With the phone on USB you can route the backend through the cable: run `adb reverse tcp:54321 tcp:54321`, then `npm run backend:local -- --host 127.0.0.1`, then restart Metro with `npx expo start -c`.

Every development build follows this computer

Any phone that once ran a development build from this computer loads its code from Metro here (port 8081). Switching `backend:local` / `backend:cloud` and restarting Metro switches all of them.

18 On an iPhone (Mac only)

You need a Mac with **Xcode** (App Store) and **CocoaPods** (`brew install cocoapods`). Then, in `apps/mobile`:

Terminal (apps/mobile)

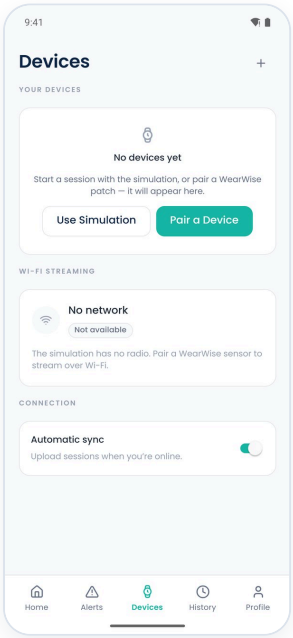
```
$ npx expo run:ios              # the iOS simulator
$ npx expo run:ios --device     # a connected iPhone
```

A real iPhone also needs your Apple developer team chosen in Xcode. See `docs/deployment.md` (section *iOS: macOS with Xcode*). The iOS app has had less real-device testing than Android.

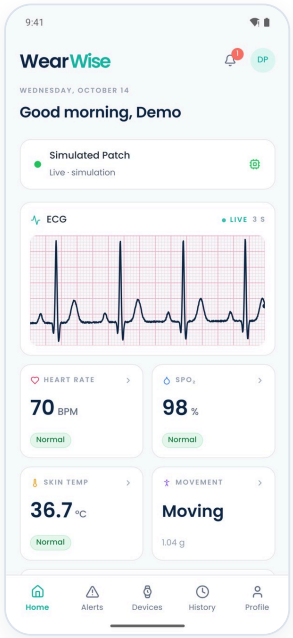
19 Try the whole system

This walk-through uses every part: a new participant records with the simulated patch and shares with the demo researcher.

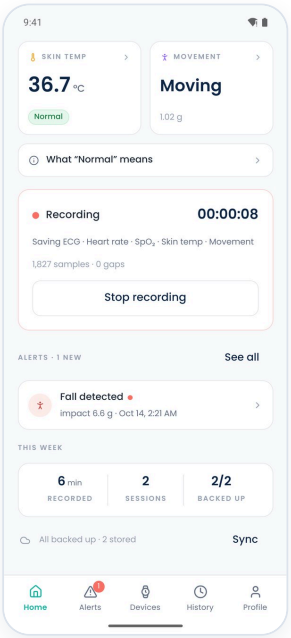
1. **Create a test participant:** in the app, sign out if needed, tap **Create Account**, and use any address (for example `you+test1@example.com`), a password of 10+ characters, and tick the three boxes. You're signed straight in.
2. **Connect the simulated patch: Devices → Use Simulation**. Home starts showing live readings. About 12 seconds later a practice **possible fall** alarm appears: tap **I'm OK**.
3. **Record:** on Home, under **Record**, tap **Start recording**, wait a minute, then **Stop recording** → **Stop**. It appears in **History** and is backed up within a minute.
4. **Check the backend:** in Studio, **Table Editor → recorded_sessions** now has your session.
5. **Share: Profile → Privacy & access → Share my data → Create code**. Note the code.
6. **Add the participant on the dashboard:** signed in as the demo researcher, click **+ Add participant** and type the code. Your test participant appears. Open it to see the session.



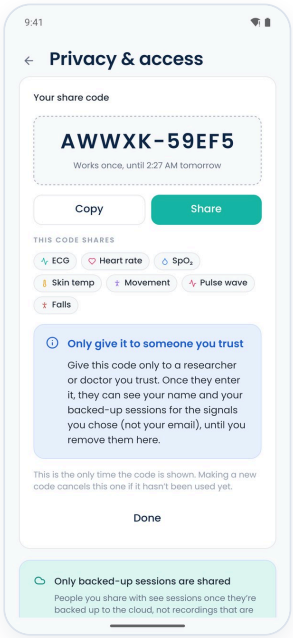
Devices → Use Simulation.



Live readings.



Recording.



The share code.

Offline test (optional)

Turn on airplane mode, or run `supabase stop`, and record a session: it records normally and waits as "not backed up". Turn the network back on (or `supabase start`) and bring the app to the front: it uploads within a minute. **Profile → Privacy & access** shows the backup status and has **Back up now**.

PART 5

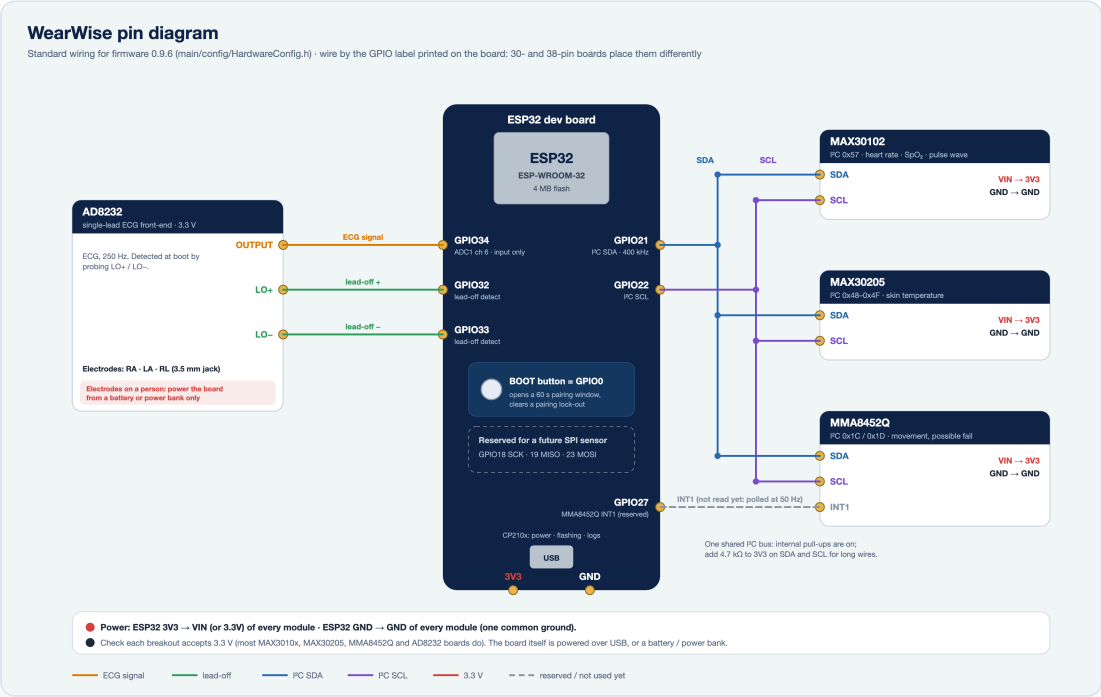
The WearWise patch

Wire the sensors, install the firmware from the dashboard, and pair the patch with a phone.

20 Parts and wiring

Part	Measures	Connects with
ESP32 dev board	Controller, Bluetooth, Wi-Fi	Classic ESP32 (ESP-WROOM-32, not S2/S3/C3), 4 MB flash, USB (CP210x or CH340 chip)
MAX30102	Heart rate, SpO_2, pulse wave	I²C (SDA/SCL)
MAX30205	Skin temperature	I²C (SDA/SCL)
MMA8452Q	Movement, possible falls	I²C (SDA/SCL); INT1 reserved
AD8232 + electrodes	ECG (single lead, education only)	Analog output plus two lead-off pins

Any subset of the sensors works: the patch reports only what it finds when it starts.



Wiring. Every module runs on **3.3 V** from the board's 3V3 pin, with all grounds connected. The three I²C modules share SDA (GPIO21) and SCL (GPIO22). Full table: `docs/hardware.md`.

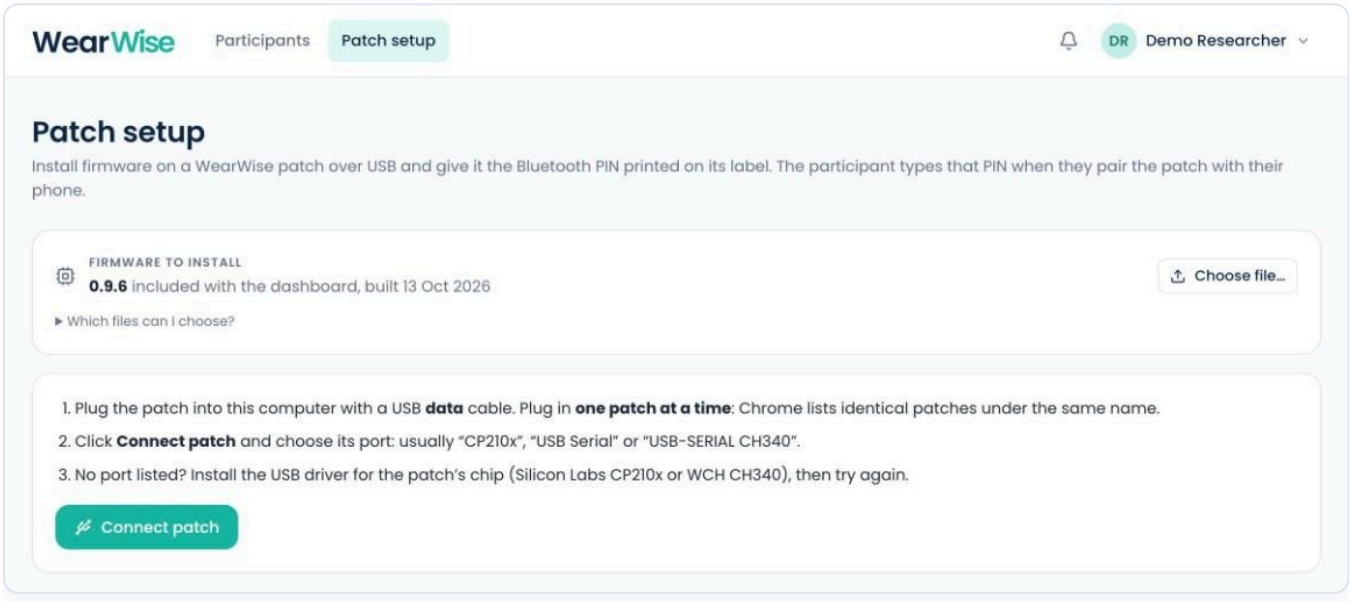
Signal	ESP32 pin	Notes
I²C SDA / SCL	**GPIO21** / **GPIO22**	MAX30102, MAX30205 and MMA8452Q. Add 4.7 kΩ pull-ups for long wires.
AD8232 OUTPUT	**GPIO34**	ECG input, sampled at 250 Hz.
AD8232 LO+ / LO-	**GPIO32** / **GPIO33**	Lead-off detection; also how the patch finds the AD8232 at start-up.
MMA8452Q INT1	GPIO27	Reserved for future use.
BOOT button	GPIO0	On the board. Opens a 60-second pairing window and clears a pairing lock-out.

Safety: ECG electrodes

Whenever ECG electrodes are on a person, power the board from a **battery or USB power bank only**, never from a mains-powered computer, charger or bench supply. The AD8232 module isn't isolated. WearWise is a research prototype, not a medical device.

21 Install the firmware (Patch setup)

The easiest way needs no programming tools: the dashboard's **Patch setup** page installs the bundled firmware (0.9.6) over USB and gives the patch its Bluetooth PIN. It works in **Chrome or Edge** on a computer, signed in as a **researcher or clinician**. Nothing about the patch is sent to the backend.



Patch setup in the dashboard's top bar.

1. Plug the patch into the computer with a USB **data** cable (some cables only charge). Plug in **one patch at a time**.

2. Click **Connect patch** and choose its port in Chrome's list: usually *CP210x*, *USB Serial* or *USB-SERIAL CH340*. No port listed? Install the USB driver for the board's chip (**Silicon Labs CP210x** or **WCH CH340**) and try again.

3. The page shows the patch's Bluetooth name (`WearWise-XXXX`), its firmware version and whether it has a label PIN. Choose:

Set up	For a **new patch**: installs the firmware, writes a new random PIN and erases the patch's settings (paired phones and Wi-Fi).
Update	New firmware, keeps the PIN, paired phones and Wi-Fi.
New PIN	For a **new participant**: a new PIN and erased settings, firmware unchanged.
Erase everything	Press and hold for 2 seconds. Wipes the whole patch. **Set up** can follow straight away.

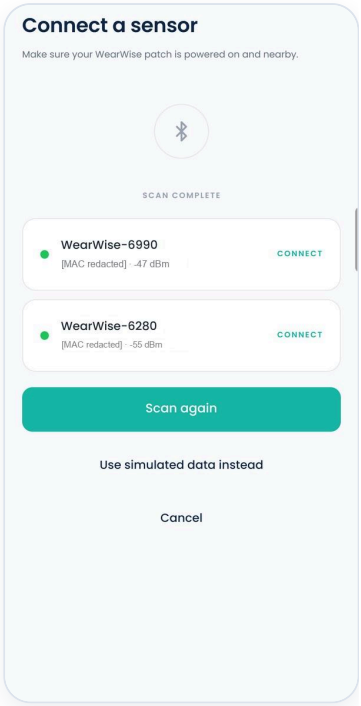
4. **Print the PIN on a label and stick it on the patch.** The participant types it when pairing.

> **"The patch didn't answer"?**
> Hold **BOOT**, tap **EN/RST**, release BOOT, and connect again. If it still fails, unplug the patch and plug it back in. *Technical details* under the card shows the full log.

Developers can also build and flash the firmware from source with ESP-IDF 5.4.2. See section 23.

22 Pair the patch with a phone

1. Power the patch (USB power bank or battery) and keep it near the phone.
2. In the app: **Devices → Pair a Device**. Allow *Nearby devices* / Bluetooth if asked.
3. Tap **CONNECT** next to the patch's name (on its label).
4. Type the **6-digit label PIN** when the phone asks. Ignore Android's "0000 or 1234" hint.
5. Home starts showing the signals the patch has. Sensors it lacks say so.



Two patches found by a real phone.

Pairing rules (enforced by the patch)

- A patch accepts a new phone only when it has never been paired, or within **60 s of pressing BOOT**.
- **3 failed attempts** lock pairing until BOOT is pressed or a pairing succeeds.
- It remembers **two** phones. A third replaces the oldest (press BOOT first). One phone is connected at a time.

Refused? In the phone's Bluetooth settings, remove the old WearWise entry, press BOOT, and pair again.

23 Optional: build the firmware from source

Only needed to change the firmware. Install **ESP-IDF v5.4.2** (Espressif's toolkit) as described in `docs/firmware.md`, then:

Terminal (ESP-IDF environment)

```
$ cd firmware/wearwise-esp32
$ idf.py set-target esp32
$ idf.py build
$ idf.py -p <PORT> flash monitor      # e.g. /dev/cu.usbserial-0001 or COM5; Ctrl ] quits the monitor
```

The serial log at start-up lists every sensor found. A board flashed this way that was never set up on the Patch setup page makes up its own PIN and prints it on this log.

PART 6

Everyday use and next steps

Starting and stopping, moving to a hosted backend, tests, and fixes for common problems.

24 Daily start and stop

Start working

1. Open **Docker Desktop**.
2. Terminal 1, in wearwise: `supabase start`
3. Terminal 2, in wearwise: `npm run dev -w dashboard`
4. Start the emulator (or plug in the phone).
5. Terminal 3, in wearwise/apps/mobile: `npx expo start -c`, then open WearWise on the device.

Stop working

1. Press `Ctrl C` in terminals 2 and 3.
2. Terminal 1: `supabase stop`. Your local data is kept.
3. `supabase stop --no-backup` also deletes the local data.
4. Quit Docker Desktop if you like.

Command (in wearwise)	What it does
`supabase status`	Is the backend running, and its addresses.
`supabase db reset`	Rebuild the local database with fresh demo data.
`npm run backend:status`	Which backend each app uses.
`npm run backend:local` / `backend:cloud`	Switch the apps between your local backend and the hosted one. Then restart Metro with `-c`.
`git pull` then `npm install`	Get the latest code from the repository.

25 Moving to a hosted backend

The local backend is for development and testing. For a real study, the organisation runs WearWise on its own **Supabase Cloud** project, or on a self-hosted Supabase over HTTPS. The same code works with each, chosen only by configuration. In outline:

1. Create a Supabase project under the organisation's account.
2. `supabase login`, `supabase link --project-ref <ref>`, then `supabase db push` to apply the same migrations. **Never** load the demo data into it.
3. Set Auth, email (SMTP) and Realtime settings in its dashboard.

4 Put its URL and **publishable** key in `apps/mobile/.env`, `apps/mobile/.env.production` and `apps/dashboard/.env`, then run `npm run backend:cloud` and restart Metro.

Release builds accept only **https://** addresses. The full steps, including release builds of the app and hosting the dashboard, are in `docs/deployment.md`.

> **Keep secrets out of the apps and the repository**
> The apps need only the project URL and its publishable key. Service keys, database passwords, access tokens and signing keys stay with their owner and are never committed. See `docs/security.md`.

26 Running the tests

What	Command (in wearwise)
Phone app: types and unit tests	`npm run typecheck -w mobile` then `npm test -w mobile`
Dashboard: types and unit tests	`npm run typecheck -w dashboard` then `npm test -w dashboard`
Database security rules (**resets the local database**)	`bash supabase/sql-tests/run-on-local-stack.sh`
Database: pgTAP tests only	`supabase test db`
Hardware test procedures	See `docs/testing/hardware-test-matrix.md`

27 Troubleshooting

Problem	Fix
`supabase start`: "Cannot connect to the Docker daemon"	Docker isn't running. Open Docker Desktop and wait for *Engine running*.
`supabase start`: "port is already allocated"	Something else uses ports 54321–54327, often another Supabase project. Stop it (`supabase stop --project-id <other>`) or quit the other program.
"command not found" for git, node, docker, supabase or adb	Open a new terminal. If it persists, reinstall that tool or check the PATH settings in section 8.
The app says the backend isn't configured	`npm run backend:local`, then restart Metro with `npx expo start -c` and reload the app.
The app can't reach `http://192.168.x.x:54321`	Same Wi-Fi (not a guest network)? Firewall allows port 54321? Did the computer's address change? Run `npm run backend:local` again and restart Metro with `-c`. On the emulator, try `--host 10.0.2.2`.
Changed the backend but the app still uses the old one	The address is bundled in: `npx expo start -c`, then reload.

Problem	Fix
"Invalid login credentials" with a demo account	The database was reset without the demo data, or the app is on another backend: `supabase db reset`; `npm run backend:status`.
`npx expo run:android` fails to find the SDK or Java	Check `ANDROID_HOME` and `JAVA_HOME` (JDK 17) from section 8, in a new terminal.
`adb devices` lists nothing, or *unauthorized*	Try another cable or USB port, turn USB debugging off and on, and accept the prompt on the phone.
Dashboard: "Not configured" page	`npm run backend:local`, then restart `npm run dev -w dashboard`.
Can't see Participants or Patch setup	You're signed in as a participant. Use a researcher account (section 12).
Share code "invalid or expired"	Codes work once, last 24 hours, and a newer code replaces an unused one. Make a new code.
Patch setup: no port in Chrome's list	Install the CP210x or CH340 driver, use a data cable, use Chrome or Edge.
Pairing refused	Remove the old WearWise entry in the phone's Bluetooth settings, press BOOT, pair again within 60 s with the label PIN.
A possible-fall alarm keeps appearing with the simulation	On purpose: about 12 s after connecting and after a recording starts or stops, then every 3 minutes. Tap *I'm OK*.
A phone you didn't mean to test with now uses the local backend	It's a development build following this computer's Metro. Run `npm run backend:cloud` and restart Metro with `-c`.

Many more cases are in `docs/troubleshooting.md`.

28 Glossary

Repository (repo)	The project's folder of code, with its history, stored in Git.
Terminal / command line	A window for typing commands (section 2).
Docker / container	Software that runs other programs (here, the Supabase backend) in self-contained boxes.
Supabase	The backend platform WearWise uses: a Postgres database, logins, file storage and live messaging.
Migration	A file that creates or changes database tables and security rules. Applied in order.
Seed / demo data	Made-up accounts and simulated recordings, loaded into the local database only.

Row Level Security (RLS)	Database rules that decide who can read or change each row. WearWise's privacy is enforced here.
Publishable key / secret key	Supabase keys. The publishable key is safe inside apps. The secret key must stay private.
Metro	The bundler that sends the app its JavaScript code while you develop.
Development build	A test version of the app that loads its code from Metro on your computer.
Emulator	A virtual Android phone running on your computer. No Bluetooth.
LAN IP address	Your computer's address on your Wi-Fi network, such as 192.168.1.20.
Port	A numbered "door" on a computer: 54321 (backend API), 54323 (Studio), 54324 (Mailpit), 5192 (dashboard), 8081 (Metro).
Firmware	The program that runs on the patch's ESP32 board.

Where to go next

The **WearWise User Guide** explains every screen of the app and the dashboard. In the repository, `README.md` is the technical quick start and `docs/knowledge-transfer.md` is the full handover guide (architecture, database, security, deployment).

Made by the WearWise Project Team

Zakiul Atheef	Project Lead, Scrum Master, Technical Lead, Security Engineering Lead and Integration Lead
Hari Bethelli	Cloud & Data Engineer
Siri Bethelli	UI/UX & Data Visualisation Engineer
Thevindu Kodagoda Munasinghage	Security Engineer, Documentation Lead and QA & Testing Lead

Project Owner: Shanmuga Sundar Dhanabalan, Deputy Director, CTI, La Trobe University (original project brief and initial project context). **Development:** the WearWise Project Team, the student development team. © 2026 WearWise Project Team · La Trobe University CSE3CAP Capstone Project. Licensed under the PolyForm Noncommercial License 1.0.0: free to use, copy and change for noncommercial purposes, not for sale. See `LICENSE`, `NOTICE` and `TEAM.md` in the repository.

WearWise Setup Guide · handover release v1.0.0, October 2026. WearWise is a research prototype and not a medical device. Security practices are informed by relevant guidance; nothing about WearWise is certified.